

- e. Cell truck or lifters
- f. One liter (quart) of touch-up paint of each color and type

10.2 COOLANT

Provide sufficient de-ionized water or cooling fluid to completely fill the cooling system plus 25 percent extra.

11.0 LV VARIABLE SPEED DRIVES

11.1 GENERAL REQUIREMENTS

11.1.1 Voltage, Frequency and Transients

The equipment shall be capable of continuous operation at input voltage deviations of +/-10% and input frequency deviations of +/-5%. In addition, contactors and relays shall be able to ride through voltage dips as specified, including those experienced during motor starting. The bus voltage shall be 400 V, 3-Ph AC with frequency of 50Hz. Maximum Voltage Dip shall be 15%. The Motor Starting Duration shall be 5 sec.

11.1.2 Assembly

The equipment shall be completely factory-built, assembled, wired, and tested. When it is necessary to partially disassemble the units for transportation, adequate materials and instructions shall be provided for ease of reassembly in the field.

11.1.3 Critical Speeds

The VSD shall be programmable to prevent continuous operation within any speed band up to +/-20% of any identified rotor critical speed. The VSD system shall be designed to run through the motor critical speeds in the shortest possible time.

11.1.4 Speed Regulation

The VSD shall achieve a speed regulation of +/-1% of the maximum speed for the speed range.

11.1.5 Voltage Regulation

The VSD output voltage shall be regulated to +/-1% under all operating conditions. The total RMS voltage unbalance shall not exceed 5% at rated load.

11.1.6 Current Rating

The VSD continuous current rating, applied at the specified environmental conditions, shall be greater than the service factor rating of the motor.

11.1.7 Efficiency

The efficiency of the VSD shall not be less than 97% under full load conditions, unless accepted otherwise by the Purchaser. The power factor at the input of the VSD shall not be less than 0.9 lagging at the normal operating load unless otherwise accepted by the Purchaser.

11.1.8 Short-Circuit Withstand

The VSD system shall be capable of withstanding the thermal and dynamic stress and the transient mechanical torque that could result from a bolted short-circuit at the VSD terminals as specified. Short-Circuit Rating of VSD shall be as shown in Single Line Diagram.

11.1.9 Output Reactor/dV/dt

An output line reactor or dV/dt filter shall be provided with the VSD package if it is required to meet motor feeder cable requirements to protect a standard motor from repetitive dV/dt spikes created by the VSD. Any motor cable length limitations or cable type requirements shall also be given in the proposal.

11.1.10 Semiconductor Operation

The inverter unit may utilize diodes for non-regenerative type drive applications. The junction temperatures of the semiconductors shall not exceed its rated value under any operating condition. The voltage imposed on individual semiconductors shall not exceed 60% of its rated voltage under any load condition.

11.1.11 DC Link

The DC link capacitors of voltage source type drives shall be film-type or foil-type. Voltage source drives shall be provided with a "braking chopper" (dynamic braking).

11.2 MAJOR AUXILIARY COMPONENTS

11.2.1 General

The VSD shall be provided as a complete package as described in this specification. The VSD Supplier shall coordinate all aspects of the drive system. The VSD shall be suitable for operating any normal, high, or premium efficiency motor having a rating up to and including the rating of the drive, with no reduction in motor life expectancy. Where required, Supplier shall provide suitable Input Isolation Transformer, Input Reactor and output reactor. The input and output reactor impedance shall be as per Supplier standards.

11.2.2 Disconnects

The VSD shall include a main disconnect with an interlocked and pad lockable handle mechanism. The main disconnect shall be rated to interrupt the available short circuit current. When specified on the datasheets, an output disconnect shall be provided.

Each disconnect shall have an auxiliary contact that opens before the power contacts to shut down the drive without damage to the drive, disconnect, or motor.

11.2.3 Bypass

The VSD assembly will have a bypass circuit containing three fully rated contactors for drive input, drive output, and bypass. The bypass circuit will also contain a motor overload relay to protect the motor during bypass operation. The drive output and bypass contactors shall be electrically and mechanically interlocked to prevent both contactors from being closed at the same time. A selector switch labelled "INV / BYP" shall select the mode of operation for the start and stop controls.

11.3 NOISE LEVEL

The maximum noise level of the unit shall not exceed 85 dBA at 1 m (36 in).

11.4 HARMONICS AND EMC

11.4.1 Harmonic Immunity

The VSD shall be immune from harmonic currents and voltages on the electrical supply system below the limits in IEC 61000-2-2, Table 1, for low voltage systems and IEC 61000-2-12, Table 1, for medium voltage systems. Immunity shall include the supplied equipment operating properly at the designed ratings.

11.4.2 Electromagnetic Compatibility

All VSDs shall comply with the requirements for Electromagnetic Compatibility (EMC) as defined in IEC 61800-3 in order to ensure that the conducted emissions in both the power supply input and output of the VSD are controlled within acceptable limits. The electromagnetic disturbance generated by the VSD, and its individual components shall not exceed a level which would affect the correct operation of both radio and telecommunications equipment. The VSD shall have an adequate level of intrinsic immunity to external electromagnetic and conducted disturbance to enable it to operate as intended.

11.5 OVERLOAD

The VSD system shall be capable of enduring a 150% overload for 60 s without component degradation or failure. After 60 s at 150% overload, the VSD shall trip. The overload of the VSD shall be settable.

11.6 COMMUNICATIONS

The VSD assembly shall have a serial data port for diagnostics, indication, programming, and control. If the need for the serial data port is not specified, the VSD must be capable of having the option installed in the future if the requirements change. Serial Communication Protocol shall be Modbus TCP/IP.

11.7 FAULTS

11.7.1 Self-Protection

The VSD shall include necessary components to protect the VSD and motor against motor overload or transient load conditions, internal faults in either motor or VSD, and disturbances in the incoming AC line. The following conditions shall cause the VSD to be shutdown with the output voltage reduced to zero:

- Motor overload
- Motor 'stall' protection (failure to breakaway and accelerate)
- Instantaneous overcurrent - VSD or motor
- DC current trip for a fault condition in any of the three power module phases
- Undervoltage or single phasing on the AC incoming line
- Overtemperature of the VSD electronics from a component or ventilation failure
- Phase or ground faults on the cable or in the motor.

11.7.2 Auto Restart

The VSD shall be capable of returning the motor to operating speed upon restoration of power following an extended voltage interruption on the AC incoming line. Automatic restart shall be selectable at the VSD panel and be disabled after an adjustable (3-30 s) time.

11.8 CATCH ON THE FLY

The VSD shall be able to restart a rotating motor ("catch on the fly") without first stopping the motor. The VSD shall sense the direction and rotation speed before catching and ramping the motor to match the input direction and speed reference.

11.9 CONTROLS

11.9.1 General

For specific control requirements of each VSD in addition to touchpad, refer to the datasheet. The VSD shall have the capability to accept external wiring for control of starting and stopping the unit. The VSD shall be capable of accepting a non-isolated 4-20 mA or 0-10 VDC speed command signal.

The VSD shall include the following:

- Assignable analog output
- Status and fault monitor visible with cabinet door closed
- Drive mounted HMI keypad
- Process control interface(s) with control signal(s) and operator devices

- Output phase and ground fault protection devices
- Serial communications interface with protocol as specified in Paragraph 11.6.

The VSD shall include the following operating adjustments with range or settings:

- Acceleration time in seconds
- Deceleration time in seconds
- Voltage offset or IR compensation
- Minimum frequency in hertz
- Maximum frequency in hertz
- Current limit in percent of output current
- Volts/hertz ratio
- Voltage boost range in percent

Supplier furnished HMI (keypad) shall include the following functionality:

- A selector switch for stopping the motor from the VSD
- Motor run status indicator
- Auto/Manual selector for speed control signal
- Manual speed control
- Trip reset control
- Motor current indication
- Motor AC voltage indication
- Motor speed indication
- Volts/hertz indication
- Motor start inhibit indication

11.10 CONSTRUCTION

11.10.1 Enclosure

Placement of the VSD chassis shall be installed in an enclosure with all components. When placed in individual enclosures, the IP rating shall be IP31 minimum. All live terminals of components located within the enclosure shall be similarly protected by barriers to a degree of protection of at least IP 20.

11.10.2 Paint

Paint Type and the paint colour shall be as per project specification for Painting and Coating.

11.10.3 Access

Cable connections, component replacement, settings, and adjustments shall be easily accomplished from the front of the VSD.

11.10.4 Cooling Type

When VSDs are placed in enclosures, Supplier shall provide heat calculations to ensure adequate cooling is provided. Water cooling or chiller cooling shall not be used.

11.10.5 Air Filters

Air filters shall be of a permanent type that can be easily removed and cleaned.

11.11 SPACE HEATERS

Provide space heaters in each cabinet to eliminate condensation. Wire together all cabinets on a single circuit. Provide a circuit breaker for each circuit for overcurrent protection and disconnection means.

Space heaters shall be provided to eliminate condensation within the VSD enclosure.

A circuit breaker shall be provided for overload protection and means of disconnection. The circuit breaker shall be mounted as per Supplier Standards.

Heaters shall be rated at twice the supply voltage of 230 VAC, Single phase - External Source.

11.12 TERMINAL BLOCKS

Provide terminal blocks for Purchaser's external wiring, conveniently located, clearly numbered, and identified. The external side of each control wire terminal shall be free of Supplier wiring terminations and shall accept a minimum of two 4 mm² wires. Power and control wiring shall be as per Supplier standards with fixed mounting.

The CT Wiring type shall be screw type shorting terminals suitable for ring type lugs and fixed mounting.

10 % spare terminals shall be provided in the terminal blocks.

11.12.1 Segregation

Connection points for inputs and outputs of different voltage levels i.e. 400 VAC, 230 VAC, 110 VDC, etc. shall be segregated from each other, use separate terminal blocks and shall be separately accessible. If necessary, this design may be accomplished with terminal barriers and covers.

11.12.2 Clearance

Terminal blocks shall be located to allow as a minimum the bending space per the IEC using the cable sizes.

11.13 WIRING

All conductors shall be copper and continuous from terminal to terminal with no splicing. Where wiring is run through a metal sheet or barrier, bushings, grommets, or other mechanical protection shall be provided. Minimum Conductor Size for CT Leads, power and control signals shall be per Supplier Standards.

11.13.1 Wire Markers

All conductor ends shall have wire markers matching Supplier's drawings. Wire Marker Type shall be per Supplier Standard.

11.14 NAMEPLATES

11.14.1 General

Nameplates shall be placed on the outside of each enclosure door identifying the service by name and equipment number. All nameplates shall be of a stainless steel material and be machine engraved. The name plate shall be English and Arabic language with White background and Black letters. The text height shall be 25mm of overall lineup. The name plate size shall be followed as supplier standard. Provide one extra set of nameplates for Purchaser use.

11.14.2 Fastening

Nameplates shall be fixed by reusable, corrosion-resistant fasteners. Adhesive backing shall NOT be permitted unless approved by the Purchaser. Fastening shall be per Supplier Standard.

11.14.3 Secondary Devices

Nameplates shall identify each device on the instrument panel and other devices such as instrument transformers and secondary fuses inside the cubicle. Identify door-mounted components both externally and internally.

11.14.4 Warning and Caution Nameplates

Install warning plates or caution notices, either in a compartment or on the outside of an assembly, where dangerous situations may be created inadvertently. Applications include, but are not limited to:

- Incoming sources with potential for feedback from an alternate source
- Termination points for all wiring from an external power source

The colour of the nameplate shall be Yellow with Black Letters. The letter height and nameplate size shall be per Supplier Standards.

11.15 AUXILIARY EQUIPMENT

11.15.1 Output Contacts

The drive shall include as a minimum the quantity of form "C" contacts for each indication. Contacts shall be rated 5 A at 230 VAC and 2 A at 110 VDC. VSD Fault indication contact quantity shall be two. For motor run indication contact shall be three while VSD Mode indication contact quantity shall be one.

11.15.2 Output Analog Signals

When specified on requisition document (VSD Remote Metering), the VSD shall provide an ungrounded analog signal directly proportional to the requested parameter.

11.15.3 Control Power

The control power transformer (CPT) volt-ampere rating(s) shall be determined by the Supplier. The rating(s) shall be sufficient to supply all power requirements of relays, indicating lights, and motor space heaters (if specified) with 25% spare capacity. A separate CPT shall be supplied by the Supplier and located in the VSD to power the motor space heater. The source shall be 230 V AC.

11.16 ISOLATION TRANSFORMERS/INPUT REACTORS

11.16.1 Requirement

When an isolation transformer or input reactor is required, the provisions of this section shall apply for dry type units.

11.16.2 Duty, Rating, Shield, and Insulation

Dry-type transformers shall be rectifier duty and rated for the application. The transformer shall contain a grounded electrostatic shield between the primary and the secondary windings. The transformer shall be supplied with an enclosure suitable for mounting external to the VSD enclosure and in accordance with the environment indicated in Section 4. The transformer shall have a Class B temperature rise with a Class H insulation system.

11.16.3 Nameplates

Nameplates shall be placed on the outside of each transformer identifying the service by name and equipment number. A rating nameplate made of stainless steel shall be provided by the Supplier and attached to each transformer. The text shall be in both English and Arabic.

Nameplate shall contain minimum essential information's as listed below,

- Manufacturer Name

- Manufacturing Date
- Model
- Ratings
- Connections
- Taps
- Serial number
- Protection class
- Compliance Marks

11.17 ACCESSORIES

11.17.1 Hardware

Furnish all accessories normally used for operation, maintenance, and testing of the equipment, including:

- Special wrenches and tools.
- One of each cable type required to connect each digital device to a Windows PC.
- One liter (quart) of touch-up paint of each color and type per lineup.

11.17.2 Software

Furnish Windows PC software normally used for operation, maintenance, and testing of all digital devices, including the drive itself and all IEDs. Software shall include all available options and add-ons, including logic programming, monitoring, diagnosis, and waveform analysis.

11.17.3 Firmware

Provide a quotation line item for any subscription or other services required for Owner to receive notifications of available equipment upgrades, including firmware releases.

Four weeks prior to testing:

- Submit a list of the latest firmware versions for all digital devices, including any recommendations to use earlier versions, for Purchaser approval.
- Confirm that all software provided to Purchaser supports the agreed firmware versions.